

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Real-Time Scalable Plant Disease Classification

Using EfficientNet

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Real-Time Scalable Plant Disease Classification Using EfficientNet**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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Abstract :

Plant diseases significantly impact global agricultural productivity, accounting for approximately 20–40% of annual crop losses. Early detection is critical to minimizing these losses, yet traditional diagnosis relies on expert knowledge, which is often unavailable to farmers in remote or resource-constrained regions.

This project presents a deep learning-based system for real-time plant disease classification using the EfficientNet family of convolutional neural networks. The system is trained on a large-scale dataset of approximately 87,000 RGB leaf images spanning 22 crop species and 38 disease classes. Among the evaluated variants (EfficientNet-B0 to B7), EfficientNet-B4 is selected as the optimal model due to its superior trade-off between accuracy (95.9%) and computational efficiency.

To enable deployment on edge devices, the trained model is converted to TensorFlow Lite (TFLite) and further optimized using INT8 quantization, reducing model size and latency while maintaining accuracy. A FastAPI-based backend exposes a RESTful prediction endpoint, and a React-based frontend provides a user-friendly interface for image upload and result visualization. The system achieves an inference latency of approximately 22 milliseconds on a mid-range Android device.

While the model demonstrates high performance on benchmark datasets, real-world evaluation indicates a moderate drop in accuracy due to environmental variations such as lighting, occlusion, and background complexity. Despite these challenges, the proposed system provides a scalable, efficient, and practical solution for precision agriculture and mobile-based disease diagnostics.

1. Introduction :

1.1 Background of the Study :

Agriculture is a cornerstone of the global economy and food supply chain. However, plant diseases remain one of the primary causes of reduced crop yield and quality. The timely identification of diseases is essential to prevent large-scale agricultural losses. Traditional methods involve manual inspection by trained agronomists, which can be time-consuming, subjective, and inaccessible in many regions.

Recent advances in Artificial Intelligence (AI), particularly in Deep Learning, have enabled automated image-based disease detection systems. Convolutional Neural Networks (CNNs) have demonstrated strong performance in visual recognition tasks, making them suitable for identifying disease patterns in plant leaves.

1.2 Problem Statement :

Despite promising results in controlled environments, existing solutions face several limitations:

- Reduced accuracy on real-world images (domain shift problem)
- High computational requirements of deep learning models
- Limited deployment on mobile or low-resource devices
- Lack of offline functionality for rural environments

Thus, the challenge is to develop a system that is **accurate, efficient, scalable, and deployable on edge devices**.

1.3 Objectives of the Study :

The primary objectives are:

- To design a deep learning-based plant disease classification system
- To evaluate and compare EfficientNet variants (B0–B7)
- To optimize the model for mobile deployment using TFLite
- To develop a complete end-to-end system (backend + frontend)
- To analyze performance under both controlled and real-world conditions

1.4 Scope of the Project :

This project focuses on:

- Image-based classification of plant diseases
- Model training, evaluation, and optimization
- Deployment on mobile and web platforms

It excludes:

- Autonomous farming systems
- Real-time drone or IoT sensor integration
- Large-scale agricultural monitoring systems

2. Methodology :

2.1 System Overview :

The proposed system follows a modular pipeline:

1. Data Collection
2. Data Preprocessing
3. Data Augmentation
4. Model Training
5. Model Optimization
6. Deployment

Each module contributes to improving model accuracy, robustness, and usability.

2.2 Dataset Description :

The dataset comprises approximately 87,000 labeled images obtained from:

- Public datasets (PlantVillage)
- Additional web-scraped images

Key Characteristics:

- 22 crop species
- 38 disease categories
- RGB images with varying resolutions

Data Split:

- Training: 80%
- Validation: 10%
- Testing: 10%

2.3 Data Preprocessing :

Preprocessing ensures data consistency and quality:

- **Resizing:** Standardizes image dimensions (224×224 or 380×380)
- **Normalization:** Scales pixel values to improve training stability
- **Cleaning:** Removes noisy, blurred, or mislabeled samples

2.4 Data Augmentation :

To improve generalization and reduce overfitting:

- Random rotations ($\pm 30^\circ$)
- Horizontal/vertical flipping
- Brightness and contrast adjustments
- Random zooming
- Gaussian blur (simulating real-world noise)

2.5 Model Architecture :

EfficientNet is used due to its efficiency and performance.

Key Components:

- Compound Scaling: Simultaneously scales depth, width, and resolution
- MBConv Blocks: Efficient convolution layers
- Squeeze-and-Excitation (SE): Enhances important features

Model Selection:

EfficientNet-B4 is selected due to optimal accuracy vs efficiency.

2.6 Training Strategy :

- Transfer learning with ImageNet weights
- Adam optimizer for faster convergence
- Cosine learning rate decay
- Cross-entropy loss with label smoothing
- Early stopping to prevent overfitting

2.7 Deployment Pipeline :

- Convert trained model → TFLite
- Optimize using INT8 quantization
- Deploy backend using FastAPI
- Develop frontend using React
- Containerize using Docker

3. Tools and Technologies :

3.1 Programming Language :

Python

Python is chosen for its simplicity, readability, and vast ecosystem of machine learning libraries.

It is used for:

- Data preprocessing
- Model training
- Backend API development

3.2 Libraries and Frameworks :

NumPy

- Provides support for multi-dimensional arrays
- Enables fast numerical computations

Pandas

- Used for structured data manipulation
- Handles missing values and transformations

TensorFlow

- Core deep learning framework
- Supports GPU acceleration
- Enables model deployment

Keras

- High-level API for building neural networks
- Simplifies training and evaluation

Scikit-learn

- Used for performance evaluation
- Provides metrics like accuracy and F1-score

Matplotlib & Seaborn

- Used for visualization
- Helps analyze training performance

3.3 Deep Learning Models :

EfficientNet

- Main model used
- High accuracy with fewer parameters

ResNet-50

- Baseline comparison model
- Deeper network but less efficient

MobileNetV2

- Lightweight model
- Suitable for low-power devices

3.4 Model Optimization :

TensorFlow Lite (TFLite)

- Converts models into mobile-friendly format
- Reduces latency and size

INT8 Quantization

- Reduces precision from 32-bit to 8-bit
- Improves speed and memory efficiency

3.5 Backend Technologies :

FastAPI

- High-performance Python web framework
- Provides REST API for predictions

Uvicorn

- ASGI server
- Handles concurrent requests efficiently

3.6 Frontend Technologies :

React.js

- Used for building interactive UI
- Enables image upload and result display

3.7 Deployment Tools :

Docker

- Containerizes application
- Ensures consistent deployment

3.8 Hardware :

- GPU: NVIDIA Tesla T4
- CPU: Intel i7
- Mobile: Snapdragon-based Android device

4. Implementation :

4.1 Model Training :

- Load dataset
- Apply preprocessing and augmentation
- Train EfficientNet model
- Validate using test dataset

4.2 Model Optimization :

- Convert model to TFLite
- Apply quantization

4.3 Backend Implementation :

- FastAPI endpoint /predict
- Accepts image input
- Returns prediction and confidence

4.4 Frontend Implementation :

- Image upload interface
- Displays results

4.5 System Workflow :

1. User uploads image
2. Image sent to backend
3. Model processes image
4. Prediction returned
5. Result displayed

5. Results :

5.1 Model Performance :

- Accuracy: 95.9%
- F1 Score: 0.954

5.2 Model Comparison :

Model	Accuracy
MobileNetV2	91.6%
ResNet-50	93.8%
EfficientNet-B4	95.9%

5.3 Inference Speed :

- Mobile: 22 ms
- CPU: 31 ms
- GPU: 6 ms

5.4 Real-World Testing :

Accuracy drops to ~89% due to:

- Lighting conditions
- Background noise
- Occlusions

5.5 Key Insights :

- EfficientNet offers best trade-off
- Quantization improves performance
- Domain shift remains a challenge

6. Conclusion and Future Scope :

6.1 Conclusion :

The project successfully develops a scalable and efficient plant disease classification system. The use of EfficientNet ensures high accuracy, while TFLite optimization enables deployment on mobile devices. The system demonstrates practical applicability in real-world agricultural scenarios.

6.2 Future Scope :

- Improve dataset with field images
- Implement object detection for localization
- Use advanced architectures
- Expand crop coverage
- Integrate IoT and drone-based monitoring

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